

Error Analysis

Project: Responsible Plant Classifier

1. Error Analysis

Teach the learning cycle

Primary target: Categorize development errors using evidence

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Precision and Recall

Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

Look for: Choose the denominator that answers the question

Reference example, not the only acceptable wording: Teacher explanation: Predicted positives $TP+FP=7$; reference positives $TP+FN=4$.

Confusion Matrix

Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

Look for: Build a labelled confusion matrix from matched reference and prediction pairs

Reference example, not the only acceptable wording: Reference clear/predicted clear contains V1,V2: 2. Clear/blocked contains V3: 1. Blocked/clear contains V4,V5: 2. Blocked/blocked contains V6,V7,V8: 3. Matrix $[[2,1],[2,3]]$.

Nearest Neighbors

Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

Look for: Compute absolute distances and rank training neighbours

Reference example, not the only acceptable wording: Distances to A,B,C,D,E are 1,1,5,7,9 cm. A and B are equally far from 3 on the numeric feature; their labels do not enter the distance calculation.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A model fails 6/10 backlit and 2/20 front-lit photos. Compute error rates and design a follow-up that tests lighting rather than leaf identity.

Concept to explain

Grouping errors by observable conditions connects a metric problem to possible data, label, feature, or model causes.

Term · Meaning

Term: error slice

Meaning: subset sharing a condition

Term: failure cluster

Meaning: related mistakes with a likely cause

Term: root-cause hypothesis

Meaning: testable explanation for errors

Term: controlled repair

Meaning: one change aimed at one cause

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on error analysis and controlled follow-up. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer. Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

It counts outcome combinations but does not explain the causes of individual errors.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank evidence table

Optional Python 3 editor; no external packages or AI accounts required

End goal for the pictorial case

By the end, I can explain error analysis using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics;

schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Cluster mistakes and propose cause-specific next tests.

MODEL IDEA Grouping errors by observable conditions connects a metric problem to possible data, label, feature, or model causes.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Twelve errors are grouped: 6 low light, 3 blurred, 2 label disagreements, 1 unusual species. Largest cluster suggests a lighting representation problem, not proof of the cause.

Draw a fresh data → model → prediction → evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 Twelve errors are grouped: 6 low light, 3 blurred, 2 label disagreements, 1 unusual species.

STEP 2 Largest cluster suggests a lighting representation problem, not proof of the cause.

STEP 3 Collect or transform low-light examples, changing one factor.

STEP 4 Rerun the same evaluation and inspect other slices for regressions.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

Four errors occur among five backlit photos; two among twenty ordinary photos.

Compare rates and investigate.

Exact explanation to model

Backlit error=80%; ordinary=10%. Inspect image quality and confounding differences. Association motivates controlled testing; it does not alone prove cause.

Explain the added worked example

Error analysis begins with prediction IDs and reference labels. Inspect features, collection conditions and the labelling policy. Do not assume every surprising prediction proves the reference is wrong. Use development errors to propose a change and specify what outcome would support the explanation.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data

split visible as you reason. Earlier weeks supplied the model-building process; this week examines error analysis and controlled follow-up.

A validation audit has 12 dim-light cases with 4 errors and 18 bright-light cases with 3 errors. Rates are $4/12=1/3$ and $3/18=1/6$. Dim lighting is a useful lead, not proof of causation: camera, route or labels may also differ. A planned change uses a training-fitted brightness normalization while keeping split and classifier settings fixed.

Trace the calculation on paper before running this optional Python check. Use the stated fictional inputs and compare intermediate values. The code checks this example; your independent workbook case uses different evidence.

Key terms

Term · Meaning

Term: Error analysis

Meaning: Inspection of mistakes to propose explanations

Term: Hypothesis

Meaning: A testable proposed explanation

Term: Controlled comparison

Meaning: A comparison that changes one planned factor while holding relevant conditions fixed

Term: Regression

Meaning: A new failure introduced by a change

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Error analysis and controlled follow-up at a glance

Part · Evidence or action · Interpretation

Part: Inspect

Evidence or action: IDs and conditions

Interpretation: Propose a hypothesis

Part: Change

Evidence or action: One planned factor

Interpretation: Keep comparison fixed

Part: Evaluate

Evidence or action: All validation records

Interpretation: Report gains and regressions

After the change, the same validation groups have 2/12 and 4/18 errors. Total errors fall from 7 to 6, but bright-light errors increase from 3 to 4. Report both results. A net improvement does not mean every part improved, and repeated experiments can overfit validation.

Does the difference prove dim light causes model errors? Explain the steps or evidence behind your response.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger

claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Error Analysis

Find an error cluster before choosing a controlled follow-up.

The facts we will use

Candidate errors: bright-light photos1/8; dim-light photos3/6. Images and reference labels are fixed for this audit.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/18

Errors per 100 photos in each lighting slice

percent

bright



dim



Bar length shows the supplied value; the marked baseline is zero.

Error Analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute slice errors

Bright: 12.5%; dim: 50%. Total errors 4/14, about 28.6%.



2. Inspect dim failures

Check blur, lighting and label uncertainty instead of assuming lighting caused every error.



3. Plan controlled pair

Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

CHANGE ONE THING

Dim errors fall to 1/6 in a later same-case replay.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Compute slice errors

What happens and why: Bright: 12.5%; dim: 50%. Total errors 4/14, about 28.6%.

Step: 2. Inspect dim failures

What happens and why: Check blur, lighting and label uncertainty instead of assuming lighting caused every error.

Step: 3. Plan controlled pair

What happens and why: Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

A slice difference suggests investigation; it does not establish a causal lighting explanation by itself.

Change one thing in the pictured case

Change: Dim errors fall to 1/6 in a later same-case replay.

Prediction: The replay's dim error rate is 16.7%; it is development evidence if those cases guided the revision.

Reason: Observed improvement on reused cases is not an untouched final generalisation test.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Four errors occur among five backlit photos; two among twenty ordinary photos.

Write the question to explain: Compare rates and investigate.

Reveal after predictions: Backlit error=80%; ordinary=10%. Inspect image quality and confounding differences. Association motivates controlled testing; it does not alone prove cause.

Keep for an independent attempt: Of eight errors, six are on blurry images and two on clear images. Can you conclude blurry images have a higher error rate without group totals?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: No. Other conditions and label quality may differ; it motivates a hypothesis. Exit response: The models are evaluated on different selected records. Apply one documented review policy and compare both on the same versioned set.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: What does a confusion matrix omit? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify error analysis in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: find the rates; explain the limit. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between error analysis and hypothesis. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Summarize the changed results without hiding a regression.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Categorize development errors using evidence; Design a controlled change with a stated hypothesis. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion task: Does the difference prove dim light causes model errors?

Teacher check: Teacher explanation: No. Other conditions and label quality may differ; it motivates a hypothesis.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Error analysis and controlled follow-up. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Dim 4/12 about 33.3 percent; bright 3/18 about 16.7 percent.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

What evidence would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

Freeze the revision and test new appropriately labelled dim-light examples, retaining plant groups and comparable conditions. Changed-case check: The replay's dim error rate is 16.7%; it is development evidence if those cases guided the revision. Observed improvement on reused cases is not an untouched final generalisation test.

Changed-case reasoning: The replay's dim error rate is 16.7%; it is development evidence if those cases guided the revision. Observed improvement on reused cases is not an untouched final generalisation test.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Error Clinic: annotate every wrong prediction, cluster by condition, write hypotheses, and select one measurable next test.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR Looking closely at one surprising error is enough to diagnose the model.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Dim 4/12 about 33.3 percent; bright 3/18 about 16.7 percent.

A stronger claim makes a stronger project: The models are evaluated on different selected records. Apply one documented review policy and compare both on the same versioned set.

The new case can be copied from the example: The independent task changes the evidence. Require a fresh explanation: Total worsens 7/30 to 8/30. A improves 50 to 30 percent, B worsens 10 to 25 percent. Investigate B and keep the original as a comparison; do not claim overall improvement.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Backlit error=80%; ordinary=10%. Inspect image quality and confounding differences. Association motivates controlled testing; it does not alone prove cause.

Independent question

Of eight errors, six are on blurry images and two on clear images. Can you conclude blurry images have a higher error rate without group totals?

Expected independent answer

No. The counts identify a useful investigation, but rates require the total number of blurry and clear examples. Inspect the images and compute each group's denominator.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “More errors in a group automatically means a higher error rate.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Rates are 60% and 10%. Photograph comparable or paired leaves under controlled lighting while holding device and labels fixed, keeping identity groups out of training/test overlap. The original association

motivates the test but does not prove lighting caused every error.

Explanations for the additional workbook examples

W1 Find the rates

Calculate the original error rates by lighting condition.

Teacher explanation: Dim 4/12 about 33.3 percent; bright 3/18 about 16.7 percent.

W2 Explain the limit

Does the difference prove dim light causes model errors?

Teacher explanation: No. Other conditions and label quality may differ; it motivates a hypothesis.

W3 Track regressions

Summarize the changed results without hiding a regression.

Teacher explanation: Total errors 7 to 6; dim 4 to 2 improves, bright 3 to 4 worsens.

W4 Apply independently

Old errors are 5/10 on route A and 2/20 on B. New errors are 3/10 and 5/20. State total and group changes and a justified next step.

Teacher explanation: Total worsens 7/30 to 8/30. A improves 50 to 30 percent, B worsens 10 to 25 percent. Investigate B and keep the original as a comparison; do not claim overall improvement.

W5 Repair a misleading audit

A team removes all disputed records only from the new model report. Why is the comparison invalid?

Teacher explanation: The models are evaluated on different selected records. Apply one documented review policy and compare both on the same versioned set.

Assess independent transfer

Selected case: Old errors are 5/10 on route A and 2/20 on B. New errors are 3/10 and 5/20. State total and group changes and a justified next step.

Selected case answer

Teacher explanation: Total worsens 7/30 to 8/30. A improves 50 to 30 percent, B worsens 10 to 25 percent. Investigate B and keep the original as a comparison; do not claim overall improvement.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Create the Plant Classifier error log linking case ID, actual, predicted, confidence, condition, likely cause, and next action.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human

review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 19: Regression and Residuals. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank evidence table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use error analysis and hypothesis in full sentences. Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Total worsens 7/30 to 8/30. A improves 50 to 30 percent, B worsens 10 to 25 percent. Investigate B and keep the original as a comparison; do not claim overall improvement. Fresh retry: old groups 2/8 and 4/12; new 3/8 and 2/12. Total improves 6/20 to 5/20 while the first group regresses. Require both statements. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Regression and residuals. Carry forward the saved evidence and the distinction between development choices and justified claims.

Plan the delayed return

Suggested return points in this level: ai-4/19, ai-4/21, ai-4/24. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.